

Exercise 4 - SQL Fundamentals (JOINS, UNION, FILTERING AND AGGREGATES

NOMPUMELELO SIMANGO

DATA ANALYTICS - MARCH 2026

SQL Joins

1. **SELECT** e.EmployeeID, e.FirstName, e.LastName
e.Department, e.Salary, p.ProjectID,
p.ProjectName, p.Budget, p.Status
FROM Employees **AS** e
INNER JOIN Projects **AS** p
ON e.EmployeeID = p.EmployeeID;

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status	
1	John	Doe	IT	70000	101	AI Development	100000	Completed	
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Ongoing	
2	Alice	Smith	HR	60000	102	Employee Training	50000	Pending	
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing	
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed	
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending	

2. **SELECT** e.EmployeeID, e.FirstName, e.LastName,
e.Department, e.Salary, p.ProjectID, p.
p.ProjectName, p.Budget, p.Status
FROM Employees **AS** e
LEFT JOIN Projects **AS** p
ON e.EmployeeID = p.EmployeeID;

Employee ID	First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70 000	101	AI Development	100 000	Completed
1	John	Doe	IT	70 000	103	Cybersecurity Audit	75 000	Ongoing
2	Alice	Smith	HR	60 000	102	Employee Training	50 000	Pending
3	Bob	Johnson	Finance	75 000	104	Financial Analysis	90 000	Ongoing
4	David	Brown	IT	72 000 65 000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65 000	105	Market Expansion	65 000	Completed
6	Michael	Clark	Finance	80 000	106	Risk Management	80 000	Pending

3. **SELECT** p. ProjectID, p. ProjectName, p. Budget,
p. Status, e. EmployeeID, e. FirstName,
e. LastName, e. Department, e. Salary

FROM Employees **AS** e

RIGHT JOIN Projects **AS** p

ON e. EmployeeID = p. EmployeeID;

Project ID	Project Name	Budget	Status	Employee ID	First Name	Last Name	Dept	Salary
101	AI Development	100 000	Completed	1	John	Doe	IT	70000
103	Cybersecurity Audit	75 000	Ongoing	1	John	Doe	IT	70000
102	Employee Training	50 000	Pending	2	Alice	Smith	HR	60000
104	Financial Analysis	90 000	Ongoing	3	Bob	Johnson	Finance	75000
105	Market Expansion	65 000	Completed	5	Emma	Wilson	Sales	65000
106	Risk Management	80 000	Pending	6	Michael	Clarke	Finance	80000

4. **SELECT** e. EmployeeID, e. FirstName, e. LastName,
e. Department, e. Salary, p. ProjectID,
p. ProjectName, p. Budget, p. Status

FROM Employees **AS** e

FULL OUTER JOIN Projects **AS** p

ON e. EmployeeID = p. EmployeeID;

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Ongoing
2	Alice	Smith	HR	60000	102	Employee Training	50000	Pending
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

5. UNION & UNION ALL

5. SELECT City AS Location FROM Employees
UNION

SELECT Status FROM Projects;

Location
New York
Los Angeles
Toronto
London
Sydney
Completed
Ongoing
Pending

6. SELECT City AS Location FROM Employees
UNION ALL

SELECT Status FROM Projects;

Location
New York
Los Angeles
Toronto
London
Sydney
New York
New York
Completed
Ongoing
Pending
Ongoing
Completed
Pending

7. Filtering Statements

7. **SELECT** EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
WHERE Salary > 70 000;

EmployeeID	FirstName	LastName	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

8. **SELECT** EmployeeID, FirstName, LastName,
Department, Salary
FROM Employees
WHERE Department IN ('IT', 'Finance');

EmployeeID	FirstName	LastName	Department	Salary
1	John	Doe	IT	70 000
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000
6	Michael	Clark	Finance	80 000

9. **SELECT** ProjectID, ProjectName, Budget, Status
FROM projects
WHERE Status <> 'Completed';

Project ID	ProjectName	Budget	Status
102	AI Development	100 000	Ongoing
103	Cybersecurity Audit	75 000	Pending
104	Employee Training	90 000	Ongoing
106	Financial Analysis	80 000	Pending
	Risk Management		

10. **SELECT** ProjectID, ProjectName, Budget, Status
FROM Projects
WHERE Budget > 70000 **AND** Status <> 'Completed';

ProjectID	ProjectName	Budget	Status
103	Cybersecurity Audit	75 000	Pending
104	Financial Analysis	90 000	Ongoing
106	Risk Management	80 000	Pending

11. **SELECT** EmployeeID, FirstName, LastName,
Department, Salary, City
FROM Employees
WHERE City IN ('New York', 'Toronto')
ORDER BY Salary **DESC**;

EmployeeID	FirstName	LastName	Department	Salary	City
6	Michael	Clark	Finance	80000	New York
3	Bob	Johnson	Finance	75000	Toronto
1	John	Doe	IT	70000	New York

12 **SELECT** ~~Department~~ ^{EmployeeID}, FirstName, LastName,
 Department, Salary
FROM Employees
ORDER BY Salary **DESC**
LIMIT 3;

EmployeeID	FirstName	LastName	Department	Salary
6	Michael	Clark	Finance	80000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000

13. Aggregate Functions with GROUP BY & HAVING

13 **SELECT** Department, **SUM**(Salary) **AS**
 TotalSalary
FROM Employees
GROUP BY Department
ORDER BY ~~TotalSalary~~ **TotalSalary** **DESC;**

Department	TotalSalary
Finance	155000
IT	142000
Sales	65000
HR	60000

14. **SELECT** City, **AVG**(Salary) **AS** AverageSalary
FROM Employees
GROUP BY City
HAVING **AVG**(Salary) > 65000;

City	AverageSalary
Toronto	75000
New York	75000

15. **SELECT** Department, **COUNT**(EmployeeID) **AS** ^{Employee}Count
FROM Employee
GROUP BY Department
HAVING **COUNT**(*) > 1;

Department	EmployeeCount
IT	2
Finance	2

16. **SELECT** Status, **COUNT**(ProjectID) **AS** ProjectCount
FROM Projects
GROUP BY Status
HAVING **COUNT**(*) >= 2;

Status	ProjectCount
Completed	2
Ongoing	2
Pending	2

17. **SELECT** e.EmployeeID, e.FirstName, e.LastName
 SUM (p.Budget) **AS** TotalProjectBudget
FROM Employees **AS** e
INNER JOIN Projects **AS** p
ON e.EmployeeID = p.EmployeeID
GROUP BY e.EmployeeID, e.FirstName, e.LastName
HAVING SUM (p.Budget) > 150000;

EmployeeID	FirstName	LastName	TotalProjectBudget
1	John	Doe	175000